

FP-AIM: Floorplan-Aware Alignment and Implicit Mapping for Sparse-Ranging Micro-Quadrotors

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Abstract—Micro-quadrotors are well-suited for confined indoor environments, but their strict payload constraints often limit onboard perception to sparse range sensing and onboard odometry. Sparse range returns provide limited geometric evidence, making structural consistency difficult to maintain from onboard measurements alone. Architectural floorplans provide global structure, but using them for mapping requires aligning the local odometry frame to the floorplan and adapting to observed local deviations from the planned structure. We present **FP-AIM**, a floorplan-aware framework that uses a 3D floorplan as a shared structural reference for sparse-range local-to-world alignment and continual implicit SDF mapping. The alignment stage estimates the local-to-world transform using a particle filter over a 2D floorplan SDF, combining sparse-beam likelihoods with candidate verification and recovery. The mapping stage warm-starts an SDF field from a floorplan-derived 3D SDF and updates it online with conflict-aware supervision, preserving floorplan-consistent structure while incorporating locally observed geometry that is not represented in the floorplan. Experiments demonstrate improved alignment reliability and reconstruction quality in simulation, while a qualitative real-world deployment shows coherent reconstruction under sparse range sensing. The code will be available on the project page: <https://fpaim-ral.github.io/fpaim-ral/>

Index Terms—Micro-quadrotor, Sparse Range Sensing, Floorplan, Global Alignment, Implicit Mapping

I. INTRODUCTION

Micro-quadrotors have emerged as a promising platform for autonomous operation in narrow and cluttered indoor environments [1]–[3], where agile navigation and a small form factor are essential. Unlike Unmanned Ground Vehicles (UGVs), which are limited by ground obstacles such as stairs, and conventional Unmanned Aerial Vehicles (UAVs), which are often impractical in confined spaces due to their size and strong downwash, micro-quadrotors can operate effectively in tight indoor settings. In such environments, complex structures and frequent occlusions make reliable 3D mapping essential. However, strict size, weight, and power (SWaP) constraints often limit micro-quadrotors to extremely sparse range sensing [1], [3], making accurate and consistent indoor 3D mapping challenging. Under extremely sparse range observations, the choice of map representation becomes critical for reliable indoor mapping. Explicit representations such as occupancy grids and TSDF/ESDF voxel maps are widely used in robotics [5]–[8], but they often become fragmented or incomplete in sparsely observed regions. Implicit representations, particularly SDF, provide a continuous formulation for representing and querying geometry beyond discrete measurements [4], [9], [10]. However, under severe sparsity, implicit models

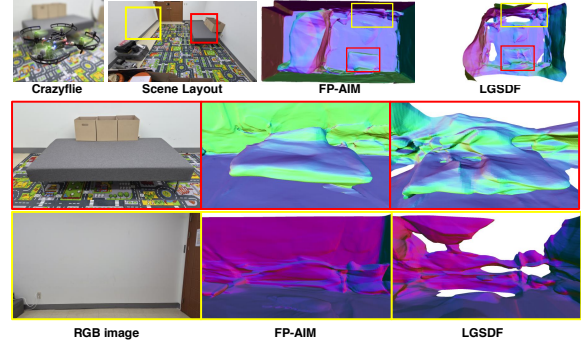


Fig. 1. Qualitative results in the real-world M09 scene. Using a Crazyflie equipped with Flow-deck local odometry and sparse Multi-ranger measurements, FP-AIM establishes a consistent floorplan-aligned frame and reconstructs coherent room-scale geometry, while LGSDf [4] exhibits fragmented surfaces and local deformation under sparse range sensing.

alone struggle to maintain global consistency and can become unstable during optimization. This suggests that representation alone is insufficient in such settings, motivating the incorporation of additional structural priors. In this work, we leverage architectural floorplans as a source of global structural information to enable more consistent reconstruction under extreme sensing sparsity [11]–[14].

Effectively leveraging floorplans under extreme sensing sparsity introduces two key challenges. First, the floorplan defines a coordinate frame independent of the robot’s local odometry frame, so the system must estimate an accurate global alignment before the mapping module can use the floorplan. Under sparse and partially occluded range observations, this alignment becomes highly ambiguous due to limited geometric constraints. Second, discrepancies between the floorplan and real-world observations are inevitable, as floorplans typically encode static structural layouts rather than the current state of the environment [11]. For example, objects or other local geometry absent from the floorplan may produce short range returns in regions marked as free space, leading to conflicting geometric constraints that can degrade reconstruction quality if not properly handled. These challenges highlight the need for a framework that can handle uncertain global alignment and floorplan–observation inconsistencies under extreme sensing sparsity.

To address these challenges, we present **FP-AIM**, a floorplan-aware framework that uses the floorplan as a shared structural reference across both global alignment and implicit SDF mapping under sparse range observations. Fig. 1 illus-

trates a representative deployment on a real Crazyflie platform. The main contributions of this work are as follows:

- We present **FP-AIM**, a floorplan-aware integrated system for sparse-ranging micro-quadrotors that uses a 3D floorplan as a shared structural reference for both global alignment and implicit SDF mapping.
- We present a floorplan-aware global alignment module that integrates sparse range likelihoods, odometry-driven particle propagation, candidate verification, and recovery into a practical lifecycle for initializing downstream floorplan-aware mapping.
- We present a floorplan-aware implicit mapping strategy that warm-starts the implicit field from a floorplan-derived structural SDF and refines it online using sparse measurements with conflict-aware hybrid supervision.
- We evaluate **FP-AIM** on simulation-generated datasets and qualitatively demonstrate on a real Crazyflie platform with onboard sparse range measurements.

II. RELATED WORK

Mapping under Sparse Sensing. Micro-quadrotors operate under stringent size, weight, and power constraints, which motivates lightweight perception and mapping from sparse range or low-power depth sensing [1]–[3]. Sparse-ranging SLAM methods exploit structural regularity, such as soft Manhattan world constraints [15] or line structures with a four-point LiDAR [16], to compensate for limited measurements. These systems are efficient, but they primarily maintain explicit or discrete representations that remain tied to directly observed regions and map resolution. In contrast, **FP-AIM** targets online implicit SDF reconstruction from sparse range observations, using a floorplan-derived structural reference to stabilize both global alignment and mapping.

Floorplan Localization and Alignment. Architectural floorplans provide compact structural references for indoor localization. Prior methods estimate poses in a floorplan frame using semantic sensor fusion [12], RGB-to-floorplan depth matching and temporal filtering [13], semantic-ray evidence [14], or floorplan-prior map construction [17]. These methods show the value of floorplans for localization, but floorplan-aware mapping also requires a stable local-to-world transform that registers the robot’s local odometry frame to the floorplan-aligned world frame. Plan-aware SLAM addresses related plan-to-observation alignment issues [11], but it targets richer SLAM settings. In contrast, **FP-AIM** estimates this transform from sparse range observations and local odometry increments, and reuses it as the structural coordinate frame for floorplan-aware implicit SDF mapping.

Online Implicit SDF Mapping. Implicit SDF maps provide continuous distance queries for reconstruction and planning. iSDF learns an online neural SDF from posed depth images [9], HIO-SDF combines hierarchical voxel structures with local neural detail [10], LGSDF improves continual ESDF learning using locally updated anchors [4], and AiSDF incorporates structural assumptions into neural SDF reconstruction [18]. These methods demonstrate the potential of

online implicit mapping, but they generally assume richer geometric input, such as dense depth, RGB-D streams, or local updates. They do not explicitly handle conflicts between architectural floorplans and sparse online range observations. **FP-AIM** addresses this setting by coupling floorplan-aware global alignment with conflict-aware implicit SDF refinement.

III. METHOD

We propose **FP-AIM**, a two-stage floorplan-aware framework for sparse-ranging micro-quadrotor mapping. Given local odometry, sparse range observations, and a 3D floorplan, **FP-AIM** first estimates the local-to-world transform and then reconstructs an implicit SDF map in the floorplan-aligned world frame. The floorplan serves as a shared structural reference across the two stages. The first stage estimates the transform by searching over horizontal translation and yaw using sparse horizontal range evidence and a horizontal floorplan SDF slice. The second stage uses the estimated transform to warm-start an implicit SDF from a floorplan-derived 3D SDF and refine it online with sparse measurements and conflict-aware supervision. The overall pipeline is shown in Fig. 2.

A. Floorplan-Aware Global Alignment

The global alignment stage estimates the rigid transform between the local odometry frame and the floorplan-aligned world frame. Given local body poses from the micro-quadrotor, the alignment module computes the local-to-world transform such that

$${}^W\mathbf{T}_B(t) = {}^W\mathbf{T}_L {}^L\mathbf{T}_B(t), \quad (1)$$

where ${}^X\mathbf{T}_Y \in SE(3)$ denotes the rigid transform from frame Y to frame X , and L , W , and B denote the local odometry, the floorplan-aligned world, and body frames, respectively. Here, ${}^W\mathbf{T}_L$ is the unknown local-to-world transform to be estimated, and ${}^L\mathbf{T}_B(t)$ and ${}^W\mathbf{T}_B(t)$ denote the body pose at time t expressed in L and W , respectively. The local odometry stream provides ${}^L\mathbf{T}_B(t)$, whereas the corresponding world-frame body pose ${}^W\mathbf{T}_B(t)$ is not directly observed. We thus estimate world-frame pose hypotheses and compute the transform from a verified pose hypothesis and the corresponding local odometry pose.

To make this inference tractable under sparse range sensing, we assume gravity-aligned local and world frames. The world frame defines the vertical z axis, while the local odometry frame shares the same vertical direction but remains offset by an unknown horizontal translation and yaw. Under this assumption, we parameterize the world-frame pose ${}^W\mathbf{T}_B(t)$ using horizontal translation and yaw while fixing the sensing height, reducing the search space to a planar 3-DoF state.

Even in this reduced search space, sparse ranges often provide insufficient information to uniquely determine the world-frame body pose. Moreover, the observed ranges may include returns from scene structures and objects that are not represented in the floorplan, introducing additional uncertainty into the alignment process. To maintain multiple hypotheses

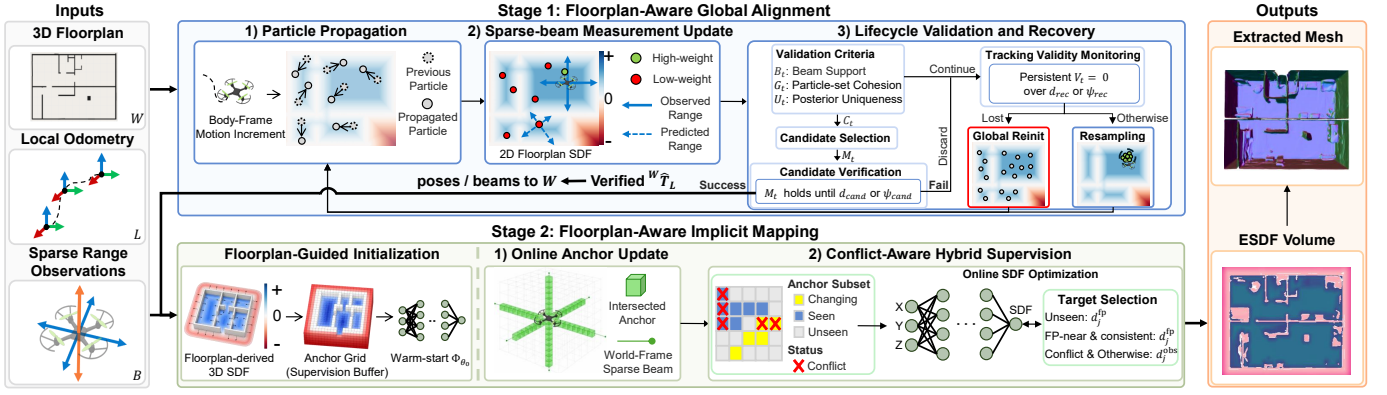


Fig. 2. Overview of FP-AIM. Stage 1 estimates the local-to-world transform from sparse range observations, local odometry increments, and a 2D floorplan SDF. Stage 2 uses the estimated transform to warm-start an implicit SDF from a floorplan-derived 3D SDF and refines it online with sparse measurements and conflict-aware supervision.

of ${}^W\mathbf{T}_B(t)$ under these uncertainties, we adopt a standard Monte Carlo localization (MCL) formulation [19], which represents the posterior belief as a weighted set of pose particles. Each particle represents a world-frame pose hypothesis, and we denote its set at time t by

$$S_t = \{\mathbf{q}_t^i\}_{i=1}^N, \quad \mathbf{q}_t^i = [\mathbf{p}_t^i, \psi_t^i]^\top, \quad (2)$$

where N is the number of particles, $\mathbf{q}_t^i$ denotes the i -th particle at time t , and $\mathbf{p}_t^i = [x_t^i, y_t^i]^\top$ and ψ_t^i denote its horizontal position and yaw, respectively.

At each time step, the alignment module updates the particle belief through propagation, sparse-beam measurement update, and resampling. The propagation step applies the motion model to the previous particle set S_{t-1} and produces the propagated set $S_t' = \{\bar{\mathbf{q}}_t^i\}_{i=1}^N$. The sparse-beam measurement update evaluates the propagated particles with the observation model and assigns normalized likelihood weights w_t^i . Resampling then forms the next particle set S_t by sampling high-weight pose hypotheses more frequently. We summarize this recursive filtering step as

$$S_{t-1} \rightarrow S_t' \rightarrow \{(\bar{\mathbf{q}}_t^i, w_t^i)\}_{i=1}^N \rightarrow S_t. \quad (3)$$

This recursion follows standard MCL, but FP-AIM specializes it for sparse-ranging floorplan-aware global alignment through three key components: planar body-frame odometry propagation, sparse-beam likelihood evaluation on a 2D floorplan SDF slice, and a validation and recovery lifecycle for transform estimation. More specifically, the lifecycle selects the highest-weight particle as a candidate pose, verifies the candidate over additional motion before computing the transform estimate, and globally reinitializes the particle set when tracking remains persistently unsupported.

Particle propagation. The motion model propagates each particle using a planar body-frame motion increment computed from consecutive local body poses and expressed in the forward-left-up convention used in ROS [20],

$$\mathbf{u}_t = [\Delta f_t \quad \Delta l_t \quad \Delta \psi_t]^\top, \quad (4)$$

where Δf_t and Δl_t denote frame-to-frame forward and left translation increments in the previous body frame, respectively, and $\Delta \psi_t$ denotes the yaw increment about the gravity-aligned vertical axis. For each particle $\mathbf{q}_{t-1}^i$, we propagate the state as

$$\bar{\mathbf{p}}_t^i = \mathbf{p}_{t-1}^i + \mathbf{R}(\psi_{t-1}^i) \left(\begin{bmatrix} \Delta f_t \\ \Delta l_t \end{bmatrix} + \boldsymbol{\eta}_t^i \right), \quad (5)$$

$$\bar{\psi}_t^i = \psi_{t-1}^i + \Delta \psi_t + \nu_t^i, \quad (6)$$

where $\bar{\mathbf{p}}_t^i$ and $\bar{\psi}_t^i$ denote the propagated position and yaw before the range-measurement update, $\mathbf{R}(\cdot)$ is the 2D rotation matrix for angle, $\boldsymbol{\eta}_t^i \sim \mathcal{N}(\mathbf{0}, \Sigma_{xy})$ is body-frame translational process noise, and $\nu_t^i \sim \mathcal{N}(0, \sigma_{\psi}^2)$ is yaw process noise.

Sparse-beam measurement update. The sparse-beam measurement update assigns an observation likelihood to each propagated particle by comparing the current horizontal range observations with ranges predicted from the horizontal floorplan SDF slice. For simplicity, we omit the time index on beam-level range quantities. For each horizontal beam k , the range sensor measures r_k along a known beam direction in B . To predict the corresponding range for $\bar{\mathbf{q}}_t^i$, we use the horizontal floorplan SDF slice extracted at the sensing height. We denote this slice by $S_{\text{FP}} : \mathbb{R}^2 \rightarrow \mathbb{R}$, where $S_{\text{FP}}(\mathbf{x})$ returns the signed distance to the nearest floorplan wall at a horizontal query location $\mathbf{x}$. We compute the predicted range as

$$\hat{r}_k^i = h_k(\bar{\mathbf{q}}_t^i; S_{\text{FP}}), \quad (7)$$

where $h_k(\bar{\mathbf{q}}_t^i; S_{\text{FP}})$ denotes SDF-based raycasting along beam k from $\bar{\mathbf{q}}_t^i$ on S_{FP} . We compute a particle weight using a Gaussian residual likelihood over valid closed-return beams,

$$w_t^i = \chi_t^i \exp \left(- \sum_{k \in \mathcal{K}_t} \frac{(r_k - \hat{r}_k^i)^2}{2\sigma_{\text{hit}}^2} \right), \quad (8)$$

where $\mathcal{K}_t$ denotes the set of valid closed-return beams within the sensing horizon at time t , excluding no-return and maximum-range beams; σ_{hit} controls the residual tolerance of the beam model; and $\chi_t^i \in \{0, 1\}$ suppresses particles outside the floorplan-valid support. We then normalize the weights

across particles such that $\sum_{i=1}^N w_t^i = 1$. The normalized weights are used for resampling and lifecycle validation.

Lifecycle validation and recovery. The sparse-beam update adjusts particle weights using current observations, but posterior concentration at a single time step does not guarantee a correct local-to-world transform. Under sparse horizontal beams, the belief can temporarily concentrate around an incorrect floorplan match. We therefore verify the candidate pose over additional motion and reinitialize the particle set only when persistent loss of support occurs.

For lifecycle validation, we evaluate three validation criteria from the current weighted particle belief, namely beam support, particle-set cohesion, and posterior uniqueness. Beam support checks whether the valid closed-return ranges agree with the ranges predicted by the maximum-weight particle. Particle-set cohesion checks whether high-weight particles are concentrated in horizontal position and yaw, while posterior uniqueness checks whether the weighted belief contains a single dominant mode rather than multiple spatially separated high-weight clusters.

To apply these validation criteria consistently, the alignment lifecycle performs candidate selection, candidate verification, and tracking-validity monitoring (see Fig. 3). The lifecycle first uses a strict condition to select a reliable candidate pose. The selected candidate is then verified using a loose condition, which checks whether the hypothesis remains supported over additional motion, thereby reducing false convergence caused by incorrect floorplan matches. Separately, tracking-validity monitoring uses a separate loose condition to determine whether the current particle belief remains sufficiently supported for continued filtering. These conditions are defined as

$$\begin{aligned} C_t &= B_t^{\text{str}} \wedge G_t^{\text{str}} \wedge U_t^{\text{str}}, \\ M_t &= B_t^{\text{loose}} \wedge G_t^{\text{loose}} \wedge U_t^{\text{loose}}, \\ V_t &= B_t^{\text{loose}} \wedge (G_t^{\text{loose}} \vee U_t^{\text{loose}}), \end{aligned} \quad (9)$$

where C_t , M_t , and V_t denote the candidate-selection, candidate-verification, and tracking-validity conditions, respectively. The terms B_t , G_t , and U_t denote the beam-support, particle-set cohesion, and posterior-uniqueness criteria. The superscripts str and loose indicate strict and relaxed thresholds, and $\wedge$ and $\vee$ denote logical conjunction and disjunction.

When the strict condition C_t is satisfied, we select the maximum-weight particle as a candidate pose and retain its pose correspondence,

$$\mathcal{P}_c = \left({}^w\hat{\mathbf{T}}_B(t_c), {}^L\mathbf{T}_B(t_c) \right) \quad (10)$$

where t_c denotes the candidate selection time. The candidate is then verified over subsequent motion. If M_t remains satisfied until the accumulated translation or yaw change after t_c exceeds its threshold, d_{cand} or ψ_{cand} , we declare convergence and use the retained correspondence $\mathcal{P}_c$ for transform estimation. If M_t fails before this motion requirement is met, we discard the candidate and continue filtering.

For recovery, the tracking-validity criterion V_t distinguishes supported tracking from persistent loss of tracking support.

When $V_t = 0$, we accumulate unsupported motion from the start time t_r of the current unsupported interval. If the accumulated unsupported translation or yaw change exceeds d_{rec} or ψ_{rec} , respectively, we replace the next particle set by global reinitialization over floorplan-valid free space. Otherwise, the next particle set is obtained by resampling from the normalized weights.

Local-to-world transform estimation. After verification, we compute the transform from $\mathcal{P}_c$ rather than from the final verification frame. This avoids perturbing the transform by particle drift during verification. We construct ${}^w\hat{\mathbf{T}}_B(t_c)$ from the candidate state using its horizontal position and yaw, the fixed sensing height, and the roll and pitch from the local odometry estimate

$${}^w\hat{\mathbf{T}}_L = {}^w\hat{\mathbf{T}}_B(t_c) ({}^L\mathbf{T}_B(t_c))^{-1}. \quad (11)$$

The transform maps subsequent local poses to the world frame for mapping.

B. Floorplan-Aware Implicit Mapping

FP-AIM enables floorplan-aware continual implicit mapping by reconciling floorplan-derived structure with sparse on-line observations through conflict-aware supervision. Whereas the original LGSDF framework builds its anchor states from accumulated sensor observations, FP-AIM introduces a floorplan-derived structural SDF as an additional source of supervision. To use this prior without forcing the reconstruction to follow the floorplan everywhere, FP-AIM maintains separate floorplan-derived and observation-accumulated states at each anchor. It then selects training targets through conflict-aware hybrid supervision, so observed deviations can override the floorplan where they disagree.

Building on the anchor-based representation of LGSDF, we use a single MLP $\Phi_\theta : \mathbb{R}^3 \rightarrow \mathbb{R}$ and a regular anchor grid $\mathcal{G} = \{\alpha_j\}_{j=1}^{N_a}$, where Φ_θ predicts the signed distance at a 3D query location, $\alpha_j \in \mathbb{R}^3$ denotes the location of the j -th anchor, and N_a is the number of anchors. The MLP represents the continuous SDF field, while the anchor grid stores local supervision used for online refinement.

The floorplan is used as a fixed structural reference during mapping. In the offline phase, we first build a floorplan-derived 3D SDF on the same regular grid used by the mapping anchors by computing the signed distance between each anchor location α_j and the 3D floorplan mesh. This 3D SDF is distinct from the horizontal 2D floorplan SDF slice used for global alignment: the 2D slice supports sparse-beam localization, whereas the 3D SDF provides structural supervision for implicit reconstruction. During online mapping, the floorplan-derived SDF values remain fixed, while the observation states, seen flags, and the neural SDF field are updated.

For each anchor, we maintain an observation-accumulated state consisting of an SDF value d_j^{obs} , observation weight w_j^{obs} , and SDF gradient $\mathbf{g}_j^{\text{obs}}$, together with a fixed floorplan-derived SDF value d_j^{fp} and a binary observation flag $m_j^{\text{seen}} \in \{0, 1\}$ indicating whether the anchor has been updated

by online measurements. The mapping process then proceeds through three steps: floorplan-guided initialization, measurement-driven online update, and conflict-aware hybrid supervision for online SDF optimization.

Floorplan-guided initialization. Before online refinement, we warm-start the neural SDF field using the fixed floorplan-derived anchor values. This provides a coherent structural initialization before online measurements are accumulated. We regress the MLP to d_j^{fp} over sampled anchors,

$$\mathcal{L}_{\text{warm}} = \frac{1}{|\mathcal{B}_{\text{warm}}|} \sum_{j \in \mathcal{B}_{\text{warm}}} \left| \Phi_{\theta}(\alpha_j) - d_j^{\text{fp}} \right|, \quad (12)$$

where $\mathcal{L}_{\text{warm}}$ is the warm-start regression loss and $\mathcal{B}_{\text{warm}}$ denotes the sampled warm-start anchor set. During this initialization, the observation states are empty and all seen flags are initialized as $m_j^{\text{seen}} = 0$. We minimize $\mathcal{L}_{\text{warm}}$ once to obtain the initial network parameters θ_0 and use them for online refinement.

Measurement-driven online update. After floorplan-guided initialization, we use online range measurements to update the observation states. At each mapping step, we process the current sparse range beams in the world frame and use all valid beams, including vertical beams, to provide 3D supervision for reconstruction. We identify candidate anchors by traversing the anchor grid along each valid beam using the Amanatides–Woo algorithm [21]. For each selected anchor, we compute a measurement-derived SDF target from the nearest valid endpoint in the current frame. The target magnitude is computed from the distance to this nearest endpoint, and the sign follows the range-based free-space convention: anchors on the free-space side of the observed surface receive positive targets, whereas anchors behind the surface within the truncation band receive negative targets.

We fuse the resulting target and gradient using the weighted accumulation rule of LGSDF. Specifically, d_j^{obs} and $\mathbf{g}_j^{\text{obs}}$ are updated by weighted averaging with the new measurement target and gradient, and w_j^{obs} accumulates the measurement weight. We then set $m_j^{\text{seen}} = 1$ for updated anchors. This update accumulates local geometry, while d_j^{fp} remains fixed as the structural reference used by the subsequent conflict-aware supervision step.

Conflict-aware hybrid supervision. We convert the stored anchor states into online SDF training targets by selecting either the fixed floorplan-derived value d_j^{fp} or the observation-accumulated value d_j^{obs} . At each mapping step, we form a training batch $\mathcal{B}_t = \mathcal{T}_{\text{chg}} \cup \mathcal{T}_{\text{seen}} \cup \mathcal{T}_{\text{unseen}}$, where $\mathcal{T}_{\text{chg}}$ contains anchors changed in the current frame, $\mathcal{T}_{\text{seen}}$ samples anchors with $m_j^{\text{seen}} = 1$, and $\mathcal{T}_{\text{unseen}}$ samples floorplan-informative anchors without online measurement support.

For seen anchors, we identify measurement-supported conflicts with the floorplan structural field as

$$m_j^{\text{conf}} = \mathbb{I} \left[w_j^{\text{obs}} \geq \tau_w \wedge d_j^{\text{fp}} - d_j^{\text{obs}} > \tau_c \right], \quad (13)$$

where $\mathbb{I}[\cdot]$ is the indicator function, $\wedge$ denotes logical conjunction, τ_w is the minimum accumulated measurement-weight

threshold, and τ_c is the conflict margin. This mask targets measurement-supported local deviations that appear closer than the floorplan-derived structural field, such as objects or other local geometry absent from the floorplan. It is therefore activated only when accumulated sparse measurements provide sufficient support for a nearer surface.

We select the SDF target for each anchor as

$$\bar{d}_j = \begin{cases} d_j^{\text{fp}}, & j \in \mathcal{T}_{\text{unseen}}, \\ d_j^{\text{fp}}, & m_j^{\text{conf}} = 0 \text{ and } |d_j^{\text{fp}}| \leq \tau_s, \\ d_j^{\text{obs}}, & \text{otherwise,} \end{cases} \quad (14)$$

where τ_s is the floorplan near-surface threshold. This rule uses floorplan supervision for unseen and floorplan-consistent near-surface anchors, and observation supervision otherwise.

We optimize the implicit field with the selected targets by minimizing the online mapping loss:

$$\mathcal{L}_{\text{map}} = \frac{1}{|\mathcal{B}_t|} \sum_{j \in \mathcal{B}_t} \rho_j \ell_{\text{sdf}}(\Phi_{\theta}(\alpha_j), \bar{d}_j) + \lambda_g \mathcal{L}_{\text{grad}}^{\text{seen}} + \lambda_e \mathcal{L}_{\text{eik}}^{\text{seen}}, \quad (15)$$

where the terms ℓ_{sdf} , $\mathcal{L}_{\text{grad}}^{\text{seen}}$, and $\mathcal{L}_{\text{eik}}^{\text{seen}}$ denote the SDF, gradient, and Eikonal losses used in neural implicit SDF mapping. In our formulation, the SDF term uses the selected target $\bar{d}_j$, while the gradient and Eikonal terms are evaluated only on seen anchors, avoiding gradient regularization in regions without online measurement support. We set ρ_j to ρ_{conf} , ρ_{seen} , or ρ_{fp} for conflict, non-conflicting seen, and unseen anchors, respectively, with $\rho_{\text{conf}} > \rho_{\text{seen}} \geq \rho_{\text{fp}}$. The optimized MLP provides the continuous SDF map, from which we query an ESDF volume and extract a mesh using Marching Cubes [22].

IV. EXPERIMENTS

A. Experimental Setup

Datasets. We utilize three OmniGibson [23] indoor scenes with different scales and object densities: `Rs_int` (small, cluttered), `Merom_0_int` (medium-scale), and `Ihlen_0_int` (large-scale, sparse-object). Sparse range observations and ground-truth poses are generated in NVIDIA Isaac Sim and replayed for all methods on an offboard server, providing identical input sequences.

Evaluation Protocol. We define three evaluation sets:

- Global-alignment set:* Three different sensing heights (`h030` / `h050` / `h100` ≈ 0.3 / 0.5 / 1.0 m) per scene, with 20 episodes each (5 for validation tuning / 15 for testing), reported over seeds 42–44 (405 test runs per method).
- Mapping-only set:* One trajectory per scene with world-aligned ground-truth poses, isolating reconstruction quality from alignment and odometry error.
- Integrated set:* One trajectory per scene linking a verified Stage 1 segment to the mapping trajectory.

For the global-alignment set and the Stage 1 segment of the integrated set, we use a drift-free synthetic local-odometry stream, isolating the local-to-world alignment from long-term odometry drift.

Baselines. For global alignment, we compare against two self-implemented particle-filter baselines. All global-alignment methods share the same floorplan SDF, odometry, range observations, maximum particle budget, sensing horizon, and likelihood form. For a fair convergence decision, the baselines use the same candidate-selection condition based on beam support, particle-set cohesion, and posterior uniqueness:

- *MCL* [19]: Classical Monte Carlo localization that applies odometry-driven particle propagation, sparse-beam likelihood weighting, and resampling, and declares convergence when the same selection condition is satisfied.
- *AMCL* [24]: An adaptive particle-filter baseline with KLD-sampling and augmented-MCL resampling components, using the same selection condition for convergence declaration.

In contrast, *FP-AIM* uses the selection condition to instantiate a candidate pose, and declares convergence only after candidate verification over accumulated motion.

For mapping, we compare against explicit and implicit SDF baselines under the same pose and observations:

- *nvblox* [8]: a GPU-accelerated volumetric ESDF mapper.
- *iSDF* [9]: an online neural SDF method trained from posed range measurements.
- *LGSDf* [4]: a continual neural ESDF method based on local anchor updates.

Evaluation Metrics. For global alignment, we report success, false-convergence, non-convergence, and failure rates, convergence frames, and pose error at convergence declaration. A trial is counted as successful when the method declares convergence with horizontal translation error ≤ 1.0 m and yaw error $\leq 30^\circ$ from the ground-truth body pose. False convergence denotes declarations outside this threshold, non-convergence denotes trials that reach the final frame before declaration, and failure is their sum. For mapping, we report SDF error, mesh accuracy, mesh completion, and Chamfer distance, with geometry extracted from a regular 256^3 grid. Mesh accuracy and completion are bidirectional nearest-neighbor distances between sampled predicted and ground-truth surfaces, with Chamfer distance their average. SDF error is evaluated at 10^6 deterministically sampled valid query points for neural methods. For *nvblox*, unknown ESDF cells are assigned zero SDF values before SDF error evaluation so that unobserved regions are included in the error computation.

Implementation Details. We emulate Crazyflie class sparse range sensing with six range beams at 20 Hz: four horizontal beams for global alignment and all six beams for mapping. All particle filter variants use a maximum particle budget of $N = 10,000$ particles and a 4.0 m sensing horizon. After validation tuning, *MCL* and *AMCL* use $\sigma_{\text{hit}} = 0.30$ m, while *FP-AIM* uses $\sigma_{\text{hit}} = 0.25$ m. For the *FP-AIM* lifecycle, candidate verification uses $d_{\text{cand}} = 0.15$ m and $\psi_{\text{cand}} = 30.0^\circ$, and recovery uses $d_{\text{rec}} = 0.3$ m and $\psi_{\text{rec}} = 100.0^\circ$.

The mapping module uses a 256^3 anchor grid. Online measurement updates follow the *LGSDf* local update rule [4] with $\eta = 5$ and $w_{\text{min}} = 10^{-3}$. Conflict-aware hybrid supervision

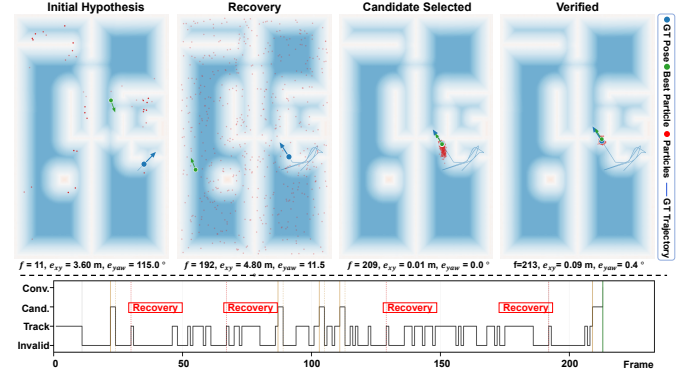


Fig. 3. Representative lifecycle of floorplan-aware global alignment. The filter initially supports an incorrect hypothesis, recovers from persistent tracking invalidity, and computes the transform estimate from a candidate that remains supported over accumulated motion. Pose errors are computed against the corresponding ground-truth poses.

uses $\tau_w = 0.5$, $\tau_c = 0.05$ m, and $\tau_s = 0.05$ m, with a weight of 2.0 for conflict anchors. All experiments run on an offboard server with an AMD Threadripper 7960X CPU and an NVIDIA RTX 4090 GPU.

B. Evaluation of Global Alignment

Tab. I reports global alignment performance over 405 held-out test runs per method under sparse observations from four horizontal beams. *FP-AIM* achieves the highest success rate, 85.19%, and the lowest failure rate, 14.81%, reducing both false-convergence and non-convergence compared with *AMCL* and *MCL*.

MCL declares convergence in fewer frames among successful trials, but it also has the highest false-convergence and non-convergence rates. *AMCL* reduces failures relative to *MCL* but requires more convergence frames. In contrast, the *FP-AIM* lifecycle uses the selection condition only to instantiate a candidate pose, verifies the candidate over accumulated motion, and reinitializes after persistent tracking invalidity. This yields fewer convergence frames than *AMCL* while achieving the lowest overall failure rate.

Fig. 3 illustrates this behavior on *Merom_0_int* at h030. The filter initially supports an incorrect hypothesis, rejects unsupported candidates, recovers from persistent tracking invalidity, and computes the transform from a candidate that remains supported through verification.

C. Evaluation of Floorplan-Aware Mapping

Tab. II evaluates mapping quality under sparse observations from six range beams. The baselines reveal the limitation of sparse range measurements alone: *nvblox* leaves large unobserved regions, reflected by higher completion error, while *iSDF* and *LGSDf* tend to interpolate smooth surfaces from sparse samples and lose local geometric details. With the floorplan as a structural reference, *FP-AIM* achieves the lowest average errors, with *FP-AIM_M*, which isolates the mapping module with world-aligned poses, reaching 2.79 cm SDF error, 10.93 cm mesh completion, and 7.44 cm Chamfer distance.

TABLE I
QUANTITATIVE RESULTS FOR GLOBAL ALIGNMENT. VALUES ARE REPORTED AS MEAN $\pm$ STANDARD DEVIATION ACROSS SEEDS. BEST RESULTS ARE SHOWN IN **BOLD**, AND SECOND-BEST RESULTS ARE UNDERLINED.

Method	Success (%) $\uparrow$	False Conv. (%) $\downarrow$	Non-conv. (%) $\downarrow$	Failure (%) $\downarrow$	Conv. Frames $\downarrow$	Trans. Err. (m) $\downarrow$	Rot. Err. (deg) $\downarrow$
AMCL	77.53 $\pm$ 0.86	15.56 $\pm$ 0.74	6.91 $\pm$ 1.13	22.47 $\pm$ 0.86	237.24 $\pm$ 10.30	0.0682 $\pm$ 0.0046	2.2027 $\pm$ 0.1458
MCL	62.22 $\pm$ 0.74	23.95 $\pm$ 2.60	13.83 $\pm$ 3.08	37.78 $\pm$ 0.74	139.37 $\pm$ 21.93	0.0662 $\pm$ 0.0055	2.2244 $\pm$ 0.0500
FP-AIM	85.19 $\pm$ 1.28	9.63 $\pm$ 0.74	5.19 $\pm$ 1.48	14.81 $\pm$ 1.28	<u>174.48 $\pm$ 22.85</u>	0.0682 $\pm$ 0.0067	1.7602 $\pm$ 0.1498

TABLE II
QUANTITATIVE 3D RECONSTRUCTION RESULTS AVERAGED OVER THREE SEEDS. AIM INDICATES FP-AIM. THE BEST AND SECOND-BEST RESULTS ARE SHOWN IN **BOLD** AND UNDERLINED, RESPECTIVELY.

Metrics	SDF Error [cm] $\downarrow$					Mesh Accuracy [cm] $\downarrow$					Mesh Completion [cm] $\downarrow$					Chamfer Distance [cm] $\downarrow$				
Methods	nvblox	iSDF	LGSDf	AIM_M	AIM_F	nvblox	iSDF	LGSDf	AIM_M	AIM_F	nvblox	iSDF	LGSDf	AIM_M	AIM_F	nvblox	iSDF	LGSDf	AIM_M	AIM_F
Rs_int	23.26	13.38	13.98	2.95	<u>2.98</u>	5.94	5.32	7.01	3.51	<u>3.61</u>	27.49	12.62	11.76	9.40	<u>9.44</u>	16.72	8.97	9.39	6.46	<u>6.53</u>
Merom_0_int	19.57	8.53	8.31	2.75	<u>3.47</u>	6.90	5.90	6.03	4.06	<u>5.24</u>	38.42	12.76	12.30	10.53	<u>10.87</u>	22.66	9.33	9.17	7.30	<u>8.06</u>
Ihlen_0_int	28.71	16.46	15.66	2.68	<u>2.92</u>	6.72	8.20	11.14	4.25	<u>4.43</u>	47.92	14.39	13.86	12.87	<u>12.95</u>	27.32	11.30	12.50	8.56	<u>8.69</u>
Average	23.85	12.79	12.65	2.79	<u>3.12</u>	6.52	6.47	8.06	3.94	<u>4.23</u>	37.94	13.26	12.64	10.93	<u>11.09</u>	22.23	9.87	10.35	7.44	<u>7.76</u>

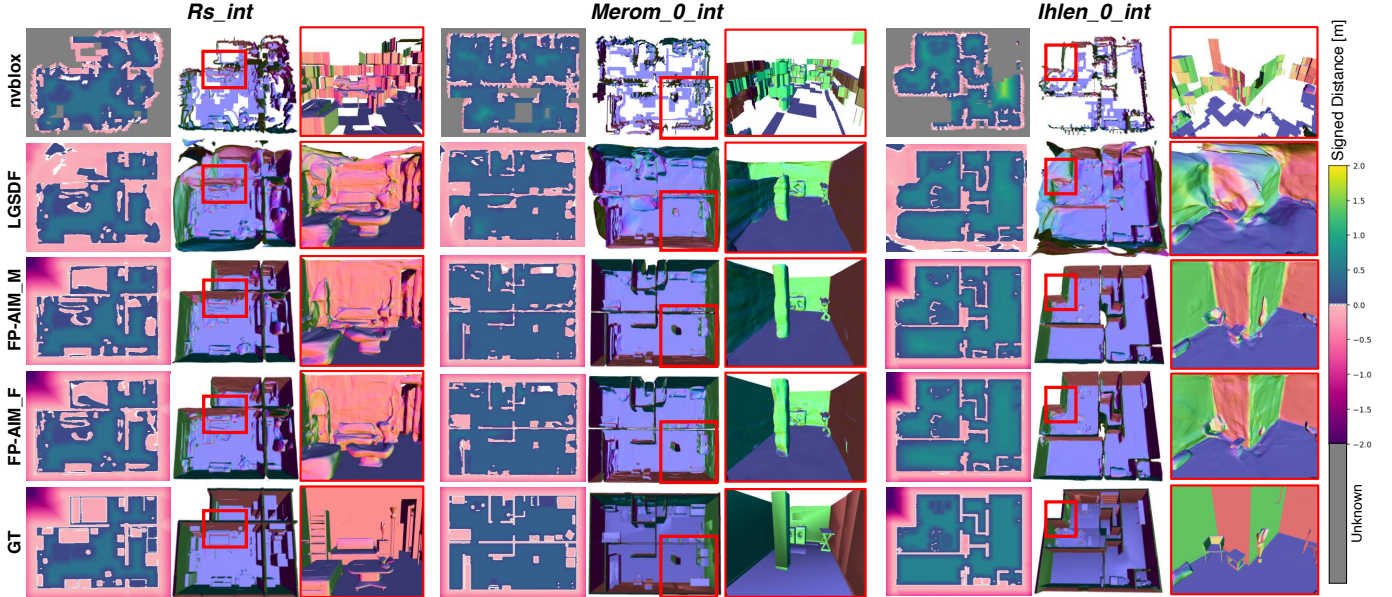


Fig. 4. Qualitative reconstruction comparison on Rs_int, Merom_0_int, and Ihlen_0_int. Red boxes mark sparsely observed wall, ceiling, and object-adjacent regions. FP-AIM preserves room-scale structural consistency and measurement-supported local deviations.

Fig. 4 shows the same trend qualitatively: FP-AIM preserves structural consistency at room scale while incorporating local deviations supported by measurements. FP-AIM_F, which uses the estimated local-to-world transform, remains close to FP-AIM_M across the averaged metrics, indicating limited degradation from global alignment.

D. Ablation Study.

Floorplan-Aware Global Alignment. Tab. III evaluates the alignment lifecycle. Removing recovery reduces the success rate from 85.19% to 55.56% and increases the failure rate to 44.44%, showing that recovery from persistent tracking invalidity is critical for reliable convergence. Removing candidate verification lowers the non-convergence rate to 0.49% while increasing false convergence to 16.79%, indicating that candidate verification suppresses incorrect convergence de-

clarations. The full lifecycle achieves the best success and failure rates by combining candidate verification over accumulated motion with recovery.

TABLE III
ABLATION STUDY OF THE FP-AIM ALIGNMENT LIFECYCLE. VALUES ARE MEAN $\pm$ STANDARD DEVIATION ACROSS SEEDS.

Variant	Succ. $\uparrow$	False Conv. $\downarrow$	Non-conv. $\downarrow$	Failure $\downarrow$
Full	85.19 $\pm$ 1.28	9.63 $\pm$ 0.74	5.19 $\pm$ 1.48	14.81 $\pm$ 1.28
w/o Recovery	55.56 $\pm$ 1.96	21.73 $\pm$ 2.60	22.72 $\pm$ 1.54	44.44 $\pm$ 1.96
w/o Cand. Verification	82.72 $\pm$ 0.43	16.79 $\pm$ 1.13	0.49 $\pm$ 0.86	17.28 $\pm$ 0.43

Floorplan-Aware Implicit Mapping. Fig. 5 ablates floorplan-guided initialization (FI), measurement-driven online update (MOU), and conflict-aware hybrid supervision (CHS) on Ihlen_0_int. FP-AIM_M achieves 2.68 cm SDF error and 8.56 cm Chamfer distance. Removing MOU raises the SDF er-

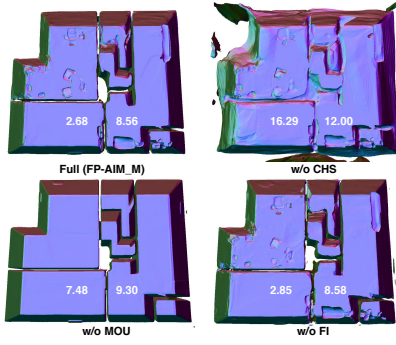


Fig. 5. Mapping ablation on *Ihlen_0_int*, averaged over three seeds. Numbers indicate SDF error / Chamfer distance in cm.

ror to 7.48 cm, confirming that online measurement updates are necessary to incorporate local geometry beyond the floorplan-derived structural prior. Removing CHS further raises it to 16.29 cm, showing that sparse online measurements must be integrated through conflict-aware target selection rather than naively overriding or following the floorplan-derived field. Removing FI has a minor effect, with 2.85 cm SDF error and 8.58 cm Chamfer distance, suggesting that FI mainly stabilizes online refinement.

E. Real-world Experiment.

We conduct a qualitative full-pipeline experiment in the M09 indoor room scene using a Crazyflie 2.1 Brushless platform with Multi-ranger and Flow decks. FP-AIM uses Flow-deck-aided EKF odometry as the local pose stream and sparse Multi-ranger measurements, while LGSDf uses the same measurements with a manually world-aligned Flow-deck pose stream. As shown in Fig. 1, FP-AIM maintains a more continuous global layout and coherent room-scale surface than LGSDf in cluttered and planar regions, consistent with the simulation trend under real sparse range sensing and Flow-deck-aided local odometry.

F. Runtime Analysis.

On the same offboard GPU server, global alignment processes each sparse range frame in 29.67 ms (33.70 Hz), exceeding the 20 Hz range stream. Online mapping refinement takes 55.70 ms per update (17.96 Hz), close to the sensor rate. These timings cover the online modules. The one-time floorplan warm-start and evaluation-only mesh extraction are outside this measurement.

V. CONCLUSION

We presented FP-AIM, a floorplan-aware framework for sparse-ranging micro-quadrotors that aligns local odometry to architectural floorplans and reconstructs implicit SDF maps in a common floorplan-aligned frame. The alignment stage improves local-to-world transform estimation through sparse-beam likelihood evaluation, candidate verification, and recovery, while the mapping stage combines a floorplan-derived structural SDF with sparse measurements through conflict-aware hybrid supervision. Simulation-generated indoor sequences show improved alignment reliability and reconstruc-

tion quality, and a qualitative real-world deployment demonstrates coherent reconstruction with onboard sparse range measurements. The current alignment stage estimates a single fixed transform, so residual yaw error can introduce floorplan-measurement mismatch during reconstruction. Future work will refine post-alignment errors, address long-term local-odometry drift and larger floorplan-scene deviations, and improve onboard resource-aware optimization.

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